

Example 1

Consider a system represented by the differential equation $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = 2r(t)$; where the initial conditions are $y(0) = 1, \frac{dy}{dt}(0) = 0$, and $r(t) = 1, t \geq 0$. What is the output $y(t)$? What is the steady state value of the output?

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = 2$$

Taking Laplace Transform on both sides,

$$\begin{aligned} \{s^2 Y(s) - sy(0) - \dot{y}(0)\} + 4\{sY(s) - y(0)\} + 3Y(s) &= 2\left(\frac{1}{s}\right) \\ \{s^2 + 4s + 3\}Y(s) &= \frac{2}{s} + s + 4 \quad \ll \gg Y(s) = \frac{2}{s(s^2 + 4s + 3)} + \frac{s + 4}{s^2 + 4s + 3} \\ Y(s) &= \frac{2 + s(s + 4)}{s(s^2 + 4s + 3)} = \frac{s^2 + 4s + 2}{s(s + 1)(s + 3)} \end{aligned}$$

Using the partial fraction

$$\begin{aligned} \frac{s^2 + 4s + 2}{s(s + 1)(s + 3)} &= \frac{A}{s} + \frac{B}{s + 1} + \frac{C}{s + 3} \\ s^2 + 4s + 2 &= A(s + 1)(s + 3) + Bs(s + 3) + Cs(s + 1) \\ \text{Put } s &= 0 \gg \gg 2 = A \times 1 \times 3 \gg \gg A = \frac{2}{3} \\ \text{Put } s &= -1 \gg \gg 1 - 4 + 2 = B \times (-1) \times 2 \gg \gg B = \frac{1}{2} \\ \text{Put } s &= -3 \gg \gg 9 - 12 + 2 = C \times (-3) \times (-2) \gg \gg C = \frac{-1}{6} \\ Y(s) &= \left(\frac{2}{3}\right)\frac{1}{s} + \left(\frac{1}{2}\right)\frac{1}{s + 1} - \left(\frac{1}{6}\right)\frac{1}{s + 3} \end{aligned}$$

Taking inverse Laplace Transform

$$y(t) = \frac{2}{3} + \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t}$$

The steady state value of the output

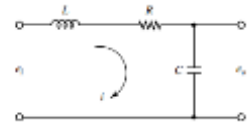
$$y_{ss} = \lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} \left\{ \frac{2}{3} + \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t} \right\} = \frac{2}{3}$$

We can also use the final value theorem,

$$y_{ss} = \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \left\{ \left(\frac{2}{3}\right)\frac{1}{s} + \left(\frac{1}{2}\right)\frac{1}{s + 1} - \left(\frac{1}{6}\right)\frac{1}{s + 3} \right\} = \frac{2}{3}$$

Example 2

Find the transfer function $G(s) = \frac{E_o(s)}{E_i(s)}$ of the R-L-C circuit shown in figure



$$e_i = e_L + e_R + e_C$$

$$e_L = L \frac{di}{dt} \quad \Leftrightarrow \quad e_R = R i \quad \Leftrightarrow \quad e_C = \frac{1}{C} \int i dt$$

Input voltage, $e_i = L \frac{di}{dt} + R i + \frac{1}{C} \int i dt$ (1)

Output voltage, $e_o = e_C = \frac{1}{C} \int i dt$ (2)

We know that

$$i = \frac{dq}{dt} \quad \Leftrightarrow \quad \frac{di}{dt} = \frac{d^2q}{dt^2} \quad \Leftrightarrow \quad \int i dt = \int \left(\frac{dq}{dt} \right) dt = q$$

$$(1) \rightarrow e_i = L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q$$

$$(2) \rightarrow e_o = e_C = \frac{1}{C} q$$

Take Laplace Transforms on both sides of eqns (1) and (2) and keeping all initial conditions equal to zero

$$E_i(s) = L s I(s) + R I(s) + \frac{I(s)}{Cs} = \left(\frac{LCs^2 + RCs + 1}{Cs} \right) I(s)$$

$$E_o(s) = \frac{I(s)}{Cs}$$

The transfer function of the system,

$$G(s) = \frac{E_o(s)}{E_i(s)} = \frac{\frac{I(s)}{Cs}}{\left(\frac{LCs^2 + RCs + 1}{Cs} \right) I(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Take Laplace Transforms on both sides of eqns (1) and (2) and keeping all initial conditions equal to zero

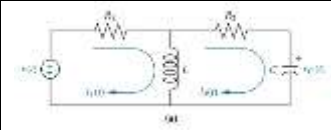
$$E_i(s) = \left(Ls^2 + Rs + \frac{1}{C} \right) Q(s)$$

$$E_o(s) = \frac{1}{C} Q(s)$$

$$G(s) = \frac{E_o(s)}{E_i(s)} = \frac{\frac{1}{C} Q(s)}{\left(Ls^2 + Rs + \frac{1}{C} \right) Q(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Example 3

Find the transfer function $G(s) = \frac{V_c(s)}{V(s)}$ of the electrical system shown



$$v(t) = i_1 R_1 + L \frac{d(i_1 - i_2)}{dt}$$

$$i_2 R_2 + \frac{1}{C} \int i_2 dt - L \frac{d(i_1 - i_2)}{dt} = 0$$

$$v_c = \frac{1}{C} \int i_2 dt$$

Taking Laplace Transform on both sides with zero initial conditions

$$V(s) = I_1(s) R_1 + L s \{ I_1(s) - I_2(s) \}$$

$$I_2(s) R_2 + \frac{I_2(s)}{Cs} - L s \{ I_1(s) - I_2(s) \} = 0$$

$$L s I_1(s) = I_2(s) \left\{ L s + R_2 + \frac{1}{Cs} \right\}$$

$$I_1(s) = \frac{L s + R_2 + \frac{1}{Cs}}{L s} I_2(s)$$

$$I_1(s) = \frac{LCs^2 + R_2Cs + 1}{LCs^2} I_2(s)$$

$$V_c(s) = \frac{I_2(s)}{Cs}$$

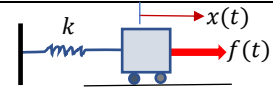
$$V(s) = I_1(s) \{ R_1 + L s \} - L s I_2(s) = \left\{ \frac{LCs^2 + R_2Cs + 1}{LCs^2} \right\} \{ R_1 + L s \} I_2(s) - L s I_2(s)$$

$$= \left[\left\{ \frac{LCs^2 + R_2Cs + 1}{LCs^2} \right\} \{ R_1 + L s \} - L s \right] I_2(s)$$

$$G(s) = \frac{V_c(s)}{V(s)} = \frac{\frac{1}{Cs}}{\left\{ \frac{LCs^2 + R_2Cs + 1}{LCs^2} \right\} \{ R_1 + L s \} - L s}$$

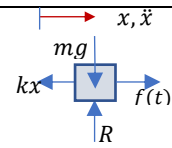
Example 4

Find the transfer function $G(s) = \frac{X(s)}{F(s)}$ of the mechanical system shown.



Consider the freebody diagram of the system when the mass moves a distance x to right of the equilibrium position.

The spring force, $F_s = kx$, where x is the relative displacement between the ends of the spring.



The equation of motion of the system is $m\ddot{x} = \Sigma F_x$

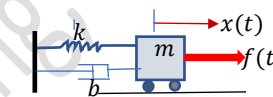
$$m\ddot{x} = f(t) - kx \ggg m\ddot{x} + kx = f(t)$$

Taking Laplace transforms on both sides and assuming zero initial conditions,

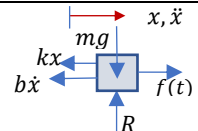
$$ms^2X(s) + kX(s) = F(s) \ggg X(s)\{ms^2 + k\} = F(s) \ggg G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + k}$$

Example 5

Find the transfer function $G(s) = \frac{X(s)}{F(s)}$ of the mechanical system represented by the spring-mass-damper shown.



The damping force, $F_d = b\dot{x}$, where b is the damping coefficient and $\dot{x}$ is the relative velocity between the ends of the damper.



The equation of motion of the system is $m\ddot{x} = \Sigma F_x$

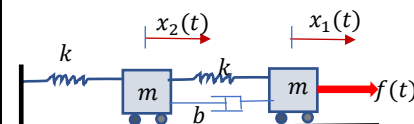
$$m\ddot{x} = f(t) - kx - b\dot{x} \ggg m\ddot{x} + b\dot{x} + kx = f(t)$$

Taking Laplace transforms on both sides and assuming zero initial conditions,

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \llgg X(s)\{ms^2 + bs + k\} = F(s) \gg G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

Example 6

Find the transfer function $G(s) = \frac{X_2(s)}{F(s)}$ of the mechanical system represented by the spring-mass-damper shown.



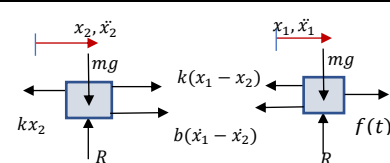
The equations of motion of the system

$$m\ddot{x}_1 = f(t) - k(x_1 - x_2) - b(\dot{x}_1 - \dot{x}_2)$$

$$m\ddot{x}_1 + b(\dot{x}_1 - \dot{x}_2) + k(x_1 - x_2) = f(t)$$

$$m\ddot{x}_2 = k(x_1 - x_2) + b(\dot{x}_1 - \dot{x}_2) - kx_2$$

$$m\ddot{x}_2 - b(\dot{x}_1 - \dot{x}_2) - k(x_1 - x_2) + kx_2 = 0$$



Taking Laplace transforms on both sides of these equations and assuming zero initial conditions,

$$ms^2X_1(s) + bs\{X_1(s) - X_2(s)\} + k\{X_1(s) - X_2(s)\} = F(s)$$

$$X_1(s)\{ms^2 + bs + k\} - X_2(s)\{bs + k\} = F(s)$$

$$ms^2X_2(s) - bs\{X_1(s) - X_2(s)\} - k\{X_1(s) - X_2(s)\} + kX_2(s) = 0$$

$$X_2(s)\{ms^2 + bs + 2k\} - X_1(s)\{bs + k\} = 0 \ggg X_1(s) = \left\{ \frac{ms^2 + bs + 2k}{bs + k} \right\} X_2(s)$$

$$X_2(s) \left\{ \frac{ms^2 + bs + 2k}{bs + k} \right\} \{ms^2 + bs + k\} - X_2(s)\{bs + k\} = F(s)$$

$$G(s) = \frac{X_2(s)}{F(s)} = \frac{bs + k}{(ms^2 + bs + 2k)(ms^2 + bs + k) - (bs + k)^2}$$

Exercise 1

Find the transfer function, $G(s) = \frac{C(s)}{R(s)}$, corresponding to the differential equation

$$\frac{d^3 c}{dt^3} + 3 \frac{d^2 c}{dt^2} + 7 \frac{dc}{dt} + 5c = \frac{d^2 r}{dt^2} + 4 \frac{dr}{dt} + 3r$$

Exercise 2

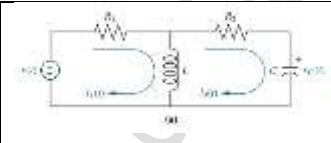
Find the differential equation corresponding to the transfer function $G(s) = \frac{2s+1}{s^2+6s+2}$

Exercise 3

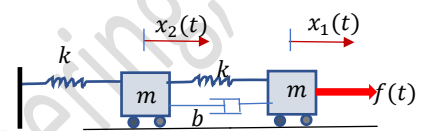
Find the ramp response for a system whose transfer function is $G(s) = \frac{s}{(s+4)(s+8)}$

Exercise 4

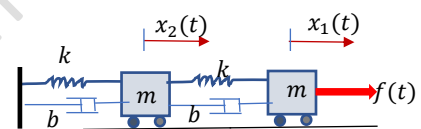
Find the transfer function $G(s) = \frac{I_2(s)}{V(s)}$ of the electrical system shown

**Example 5**

Find the transfer function $G(s) = \frac{X_1(s)}{F(s)}$ of the mechanical system represented by the spring-mass-damper shown.

**Example 6**

Find the transfer function $G(s) = \frac{X_2(s)}{F(s)}$ of the mechanical system represented by the spring-mass-damper shown.

**Example 7**

Obtain the transfer function of a dc motor i) armature controlled ii) field controlled

Example 8

Obtain the partial fraction expansion of the following function using MATLAB/SCILAB:

$$F(s) = \frac{10(s+2)(s+4)}{(s+1)(s+3)(s+5)^2}$$

Then, obtain the inverse Laplace transform of $F(s)$.

Example 9

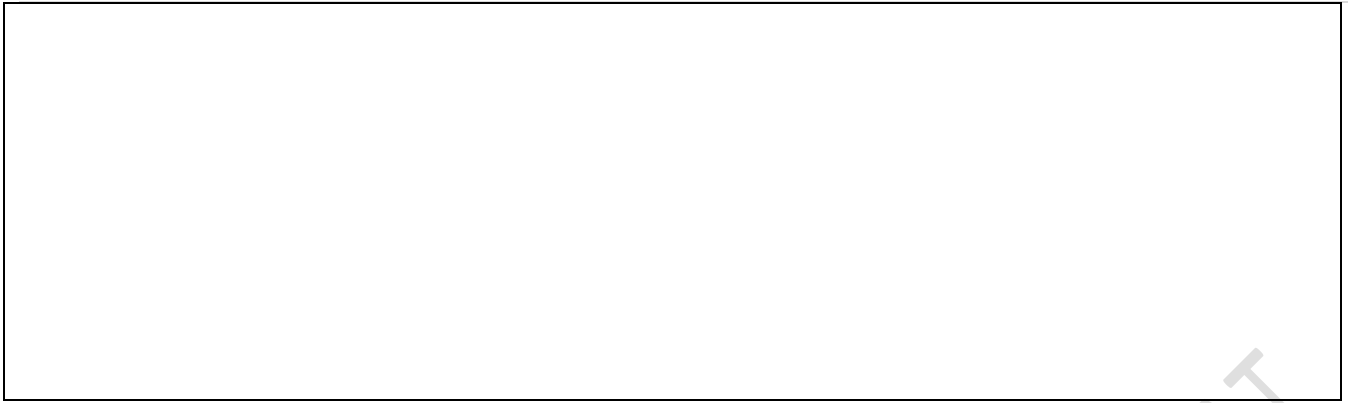
Obtain the partial fraction expansion of the following function using MATLAB/SCILAB:

$$F(s) = \frac{(s+1)}{s(s^2+s+1)}$$

Then, obtain the inverse Laplace transform of $F(s)$.

References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons



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